TEAM ASTRO

AeroSentinel

Satellite recovery decisions with protected mission history

8 task families / 3 difficulty levels / tool-using agents

A declared benchmark simulator

Each action changes a persistent mission

Layer	Agent sees or controls	Verifier evaluates
Observations	Manual, configs, mission targets and live telemetry	Seeded protected facts
Actions	Read, write, run Python, advance orbit, finish	Time, energy, exposure and delivered data
Limits	40 replies and 10 tools per reply	600 seconds plus a separate 120-second final check

The same tool and reward contract applies to every configured model.

Task depth progresses from repair to scheduling

Group	Task families	Decision required
5 guided repairs	Power, pointing, thermal, radio and fault cascade	Diagnose and repair seeded faults
1 radio mission	RF energy tradeoff	Balance data, link margin and transmitter energy
2 temporal missions	Early pass recovery Observation scheduling	Recover before contact Schedule exposure and energy

Seeds vary faults, hardware profiles, deadlines and mission requirements.

A constant payload setting cannot meet both goals

Useful exposure requires duty ≥ 0.90

Whole-mission energy permits duty ≤ 0.126–0.189

A feasible plan changes duty over time

Instrument off before the window. On during useful exposure.
Off after the window. Every switch consumes simulated time.

Reward tracks progress and mandatory mission checks

Raw component	Standard mission	Observation mission
Power / thermal	0.35 / 0.25	0.28 / 0.20
Pointing / communications	0.20 / 0.20	0.16 / 0.16
Useful exposure	Not scored	0.20

Progress = clamp((raw − no-op raw) / (1 − no-op raw), 0, 1)

Every required check must pass. Catastrophic failures remain latched.

Separate processes protect scoring and mission facts

Boundary	Authority	Protection
Harness	Verifier, reward, mission truth	Separate container Authenticated controls
Agent worker	Workspace repairs and tools	UID 1001, no outbound route Read-only root and resource limits
Behavioral evaluator	Runs candidate calculation scripts	UID 1002, bounded fixtures Worker cleanup between calls

Docker and the host kernel remain part of the trust boundary.

Executed attack probes produce honest failures

Executed probe	Reward	Termination
Relax pointing requirement	0.800	failure
Forge telemetry output	0.000	failure
Miss early contact	0.418	failure
Miss observation window	0.800	failure

Four scripted probes. No model API calls. All failed the success checks.

REPRODUCIBILITY EVIDENCE

Tests measure feasibility and shortcut resistance

71 Linux regressions **160 task/seed pairs** **12 live isolation checks**

Scripted policy	Successful episodes	Evidence scope
No-op	0 / 40	8 tasks × 5 separate seeds
Nominal / max radio	25 / 40 and 20 / 40	0 / 15 on the three hard tasks
Privileged reference	40 / 40	Feasibility baseline

All 8 gates pass. The offline policy comparison never calls a model.

Final-source model evidence includes connection failures

Attempt	Outcome / reward	Replies	Harness seconds
EPS Run A, no append	Success / 1.0	7	29.31
EPS + sample append	Success / 1.0	10	44.72
Early pass, seed 101	Connection crash / 0.0	1	12.81
Observation, seed 101	Connection crash / 0.0	0	7.14

2 successes, 2 provider failures. Hard-task model capability remains unproven.

The demo is ready, with explicit limits

Run A: EPS recovery, seed 0

Reward 1.0 **7 replies** **29.31 harness seconds** **No append**

Known limits

Simplified dynamics. Five guided repair tasks. Finite exploit coverage.
One tested deployment. Provider connectivity affected hard-task runs.

Submission still needs privacy/name-case review, organizer access and form upload.